

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05 COURSE

OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)*

Assessment Tool Weightage: 40%

QUESTIONS

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

Answer: Sampling converts a continuous scene into discrete pixels at fixed intervals. If the sampling rate is too low (below the Nyquist rate), fine spatial detail is lost and the image appears blurred, with edges poorly defined. This can be corrected by increasing the sampling rate above the Nyquist limit, or by applying an anti-aliasing (low-pass) filter before sampling and using interpolation to reconstruct missing detail.

2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

Answer: Under low light, fewer photons reach the sensor, reducing the signal-to-noise ratio. To compensate, the acquisition system may increase exposure time or sensor gain, but this amplifies electronic and thermal noise, reduces contrast, and can introduce grain/noise into the digital image, lowering overall image quality.

3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

Answer: Aliasing occurs when the image is sampled below the Nyquist rate, causing high-frequency details to be misrepresented as false low-frequency patterns; coarse quantization can add further distortion by exaggerating these errors. The corrective approach is to apply an anti-aliasing (low-pass) filter before sampling and to use an adequate sampling rate with sufficient quantization levels.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Answer: Pixel (spatial) resolution refers to the number of pixels used to represent the spatial detail of an image, while intensity resolution refers to the number of gray levels (bit depth) used to represent brightness. In medical imaging, high pixel resolution is needed to capture fine anatomical structures, while high intensity resolution is needed to distinguish subtle tissue contrast differences that are critical for accurate diagnosis.

5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Answer: Such a system involves all three levels: low-level (image acquisition and preprocessing such as noise removal), mid-level (segmentation, feature extraction, and object tracking), and high-level (decision making, such as recognizing suspicious activity). Real-time surveillance is justified as requiring all levels because raw frames must be quickly processed, analyzed, and interpreted continuously to enable timely alerts.

6. Explain how image formation models help in improving image quality in applications such as autonomous driving.

Answer: Image formation models mathematically describe how a 3D scene is projected onto a 2D image plane, accounting for camera geometry, lens distortion, and illumination. In autonomous driving, these models are used for camera calibration and distortion correction, enabling accurate depth and distance estimation, which improves the reliability of scene interpretation and decision-making.

7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Answer: Quantization converts the continuous range of intensity values into a finite number of discrete levels (bit depth). A higher number of quantization levels preserves smooth gradients and subtle intensity variations, while fewer levels cause visible contouring/banding and loss of fine intensity detail.

8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Answer: Noise originates during acquisition from sources such as sensor thermal noise, low illumination, and electronic interference, and is worsened by high gain settings used to compensate for weak signals. It can be minimized using better-quality sensors, proper exposure and illumination, cooling, averaging multiple frames, and post-acquisition noise-filtering techniques.

9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Answer: At the low level, the system acquires and preprocesses the image (noise removal, normalization). At the mid level, it extracts features such as facial landmarks and edges. At the high level, it performs recognition/classification, matching extracted features against stored data to make an identity decision.

10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Answer: Low sampling reduces spatial resolution, so fine details and edges of objects are lost or merged with the background. Errors in image formation, such as blur or distortion, further degrade clarity. Together these reduce the visibility of small or distant objects, which can be improved with denser sampling (higher resolution) and correct focus/calibration.

11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.

Answer: CCD sensors generally provide higher image quality, lower noise, and better light sensitivity, but are slower, consume more power, and are costlier to produce. CMOS sensors are faster, more power-efficient, cheaper, and allow on-chip integration, though they have traditionally shown relatively higher noise. The choice depends on whether the application prioritizes image quality (CCD) or speed, cost, and power efficiency (CMOS).

12. A digital image appears distorted due to improper image formation. Identify possible causes and suggest improvements based on image formation models.

Answer: Possible causes include lens distortion, incorrect camera calibration, perspective projection errors, and sensor misalignment. Improvements can be made through proper camera calibration, lens distortion correction algorithms, better optical design, and applying geometric image formation models to correct the distortion.

13. Explain how the features of Computer Vision systems enable applications such as automated surveillance.

Answer: Features such as object detection, motion tracking, and pattern recognition allow a system to automatically identify objects, track movement, and detect anomalies in a scene without human supervision. This enables continuous, real-time monitoring and automatic alerting in automated surveillance systems.

14. A system processes satellite images for land analysis. Explain how digital image fundamentals and acquisition methods influence the accuracy of results.

Answer: Fundamentals such as spatial and spectral resolution determine how finely land features can be distinguished, while acquisition methods (sensor type, atmospheric conditions, calibration) affect the clarity and consistency of captured data. Higher resolution and well-calibrated acquisition improve the accuracy of land classification and analysis.

15. Given an application like OCR (Optical Character Recognition), explain how sampling, quantization, and image acquisition collectively affect performance.

Answer: Sampling determines how clearly character edges and strokes are captured; inadequate sampling blurs characters. Quantization affects the contrast between text and background, influencing segmentation. Acquisition quality (lighting,

focus, sensor noise) affects the overall clarity of the input. Together, these factors determine how accurately characters can be segmented and recognized.

16. A high-resolution image consumes excessive storage and processing time. Explain how sampling and quantization can be adjusted to optimize performance without significant quality loss.

Answer: The sampling rate can be reduced (downsampling) to decrease the number of pixels, and the number of quantization levels can be reduced moderately, as long as both remain above the threshold of noticeable perceptual quality loss. Combined with compression techniques, this balances storage/processing efficiency with acceptable image quality.

17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.

Answer: Image formation depends on the interaction between illumination and scene reflectance; uneven lighting causes shadows and non-uniform captured radiance, resulting in poor contrast. Corrective measures include histogram equalization, adaptive illumination correction, and using the image formation model to separate and normalize the illumination component.

18. In a medical imaging system, explain how improper quantization can lead to loss of critical diagnostic information.

Answer: If too few intensity levels are used, subtle contrast differences between healthy and abnormal tissue (e.g., a small tumor) may be merged into the same gray level, causing loss of important diagnostic detail. Using a higher bit-depth quantization preserves these subtle variations, aiding accurate diagnosis.

19. A self-driving car system relies on visual input. Explain how image acquisition and Computer Vision features contribute to reliable decision making.

Answer: Acquisition (cameras and sensors) provides continuous, real-time raw visual data of the surrounding environment. Computer Vision features such as object detection, segmentation, and tracking process this data to identify road elements, pedestrians, and obstacles, enabling the system to make reliable, informed driving decisions.

20. An application requires detecting fine details in images. Explain how sampling rate and intensity resolution should be chosen to meet this requirement.

Answer: A high sampling rate is needed to capture small spatial variations without losing detail, and a high intensity resolution (more gray levels) is needed to distinguish subtle differences in brightness or contrast. Both must be sufficiently high together, since improving only one is insufficient for fine detail detection.

21. An image captured at a low sampling rate shows jagged edges. Explain how sampling rate affects spatial resolution and suggest improvements.

Answer: A low sampling rate reduces spatial resolution, causing curved or diagonal edges to be represented as a staircase-like pattern of jagged pixels (aliasing along boundaries). This can be improved by increasing the sampling rate, applying anti-aliasing filters, or using interpolation to smooth the edges.

22. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.

Answer: Banding occurs when too few quantization levels are used to represent a smooth intensity gradient, creating visible discrete steps instead of a smooth transition. The solution is to increase the number of quantization levels (higher bit depth) or apply dithering to visually smooth the transitions.

23. In a thermal imaging system, explain how image sensing affects temperature-based visualization accuracy.

Answer: The sensor's sensitivity to infrared radiation, its resolution, and its noise level determine how accurately temperature differences in the scene are captured and mapped to intensity values. Poor sensing quality can lead to inaccurate or noisy temperature visualization and false readings.

24. A blurred image is obtained due to motion during acquisition. Explain how image formation and sensing contribute to this effect.

Answer: During acquisition, if the scene or camera moves while the sensor is exposed, the image formation process integrates light over the exposure time, causing the moving object's radiance to smear across multiple pixel positions, producing motion blur. This can be reduced with faster shutter speeds, stabilization, or deblurring algorithms.

25. A satellite image shows loss of fine terrain details. Explain how sampling and resolution affect feature visibility.

Answer: A low sampling rate/resolution means each pixel represents a larger ground area, merging fine terrain features into fewer pixels and losing detail. Higher-resolution imaging systems with denser sampling are needed to preserve the visibility of small terrain features.

26. In a biometric system, explain how image acquisition quality affects recognition accuracy.

Answer: Poor acquisition, such as inadequate lighting, sensor noise, or low resolution, reduces the clarity of biometric features (e.g., fingerprint ridges or facial landmarks), making feature matching less reliable. High-quality acquisition preserves fine detail and improves recognition accuracy.

27. An image appears pixelated when zoomed. Analyze the role of sampling and resolution in this effect.

Answer: Low sampling/resolution means the image is represented by relatively few pixels. When the image is zoomed in, each pixel is enlarged and becomes visibly

blocky, producing the pixelated appearance, since there is insufficient sampled detail to fill in the enlarged view smoothly.

28. A digital image shows reduced contrast due to poor sensing conditions. Explain how acquisition parameters influence this.

Answer: Acquisition parameters such as exposure time, sensor gain, and sensitivity determine how much of the scene's dynamic range is captured. Under poor lighting or incorrect settings, the captured intensity range is compressed, resulting in reduced contrast in the final image.

29. In a machine vision system, explain how improper quantization affects defect detection.

Answer: Coarse quantization merges subtle intensity variations that indicate small surface defects into the same gray level, causing them to go undetected. Finer quantization preserves the intensity contrast associated with defects, improving detection reliability.

30. A camera captures images with varying brightness levels. Explain how image formation models influence intensity variations.

Answer: Image formation models describe intensity as a function of illumination and surface reflectance. Variations in light source intensity, angle, or scene reflectance properties directly change the captured brightness, and these models help explain and correct for such intensity variations across the image.

31. An image processing system fails to detect edges clearly. Explain how sampling and acquisition affect edge visibility.

Answer: Insufficient sampling (low resolution) and acquisition issues like noise or blur smooth out the sharp intensity transitions that define edges, making them harder to detect. Higher resolution sampling and noise reduction during acquisition improve edge visibility for detection algorithms.

32. In an industrial inspection system, explain how image sensing impacts defect detection reliability.

Answer: The resolution, illumination, and sensor type used during sensing determine how well subtle surface defects are captured. Poor sensing quality reduces the contrast and detail of defects in the captured image, lowering the reliability of automated defect detection.

33. A low-bit image loses important details. Explain how quantization affects information preservation.

Answer: Using fewer bits per pixel means fewer available intensity levels, so multiple distinct original intensity values get mapped to the same level (quantization error), causing loss of subtle detail. Increasing bit depth preserves more of the original intensity information.

34. A surveillance image appears noisy in low light. Analyze how sensing and acquisition conditions contribute to noise.

Answer: In low light, the signal captured by the sensor is weak relative to its inherent noise floor, resulting in a low signal-to-noise ratio. Compensating with higher gain during acquisition amplifies this noise further, producing a visibly grainy/noisy image.

35. A digital image suffers from aliasing when resized. Explain how sampling principles apply in this scenario.

Answer: Resizing changes the effective sampling rate of the image. If downsampling is performed without first applying a proper low-pass (anti-aliasing) filter, the new sampling rate violates the Nyquist criterion for the image's frequency content, producing aliasing artifacts such as moiré patterns.

36. In medical imaging, explain how increasing sampling rate improves diagnostic accuracy.

Answer: A higher spatial sampling rate captures finer anatomical detail, increasing the ability to resolve small structures or abnormalities that might otherwise be missed at coarser sampling, thereby improving the accuracy of diagnosis.

37. A system captures images with inconsistent quality. Explain how acquisition methods influence uniformity.

Answer: Variations in sensor calibration, illumination conditions, or capture settings between sessions lead to inconsistent image quality across captures. Using standardized, well-calibrated acquisition protocols helps maintain uniform image quality.

38. A high-speed camera captures distorted images. Explain how image formation and sampling contribute to distortion.

Answer: At high speeds, motion during the limited exposure time combined with the image formation process can cause motion blur and rolling-shutter type distortion, while limited spatial sampling further reduces the fidelity with which fast-moving details are captured, adding to the distortion.

39. In an OCR system, explain how poor sampling affects character recognition accuracy.

Answer: Low sampling blurs and loses fine strokes and edges of characters, making character shapes ambiguous or merging adjacent characters. This reduces the accuracy of character segmentation and recognition.

40. A digital image shows quantization noise. Explain its origin and suggest mitigation techniques.

Answer: Quantization noise arises from the rounding error introduced when continuous intensity values are mapped to a finite set of discrete levels. It can be mitigated by using a higher bit depth (more quantization levels), dithering, or noise-shaping techniques.

41. A vision system processes images with varying illumination. Explain how image formation models help normalize such variations.

Answer: Image formation models separate the captured intensity into illumination and reflectance components. By modeling and estimating the illumination component, the system can normalize it out, producing a more consistent representation of the scene's true reflectance regardless of lighting changes.

42. An image appears overly smooth with loss of detail. Explain how quantization levels contribute to this issue.

Answer: When too few quantization levels are used, subtle intensity gradations and textures are merged into broader flat regions, making the image appear overly smooth and losing fine detail and texture information.

43. In a facial recognition system, explain how resolution impacts feature extraction accuracy.

Answer: Higher resolution captures finer facial features such as landmarks, textures, and small distinguishing details, improving the accuracy of feature extraction. Low resolution loses these discriminative details, reducing the accuracy of subsequent recognition.

44. A camera system produces images with color distortion. Explain how sensing and acquisition affect color representation.

Answer: Color representation depends on the sensor's color filter response, white balance calibration, and the color temperature of the illumination during acquisition. Miscalibration of any of these acquisition parameters causes inaccurate color reproduction, resulting in color distortion.

45. A digital image has insufficient contrast. Explain how image acquisition and formation influence contrast levels.

Answer: Acquisition settings such as exposure and illumination determine how much of the scene's dynamic range is captured, while the image formation process determines how reflectance and lighting combine into intensity values. Poor lighting or incorrect exposure during acquisition compresses the captured range, producing insufficient contrast.

46. A system requires detecting small objects in images. Explain how sampling and resolution should be adjusted.

Answer: A higher spatial sampling rate (higher resolution) should be used so that small objects are represented by enough pixels to be distinguishable from the background and noise, rather than being lost between sparse samples.

47. In satellite imaging, explain how quantization affects classification accuracy.

Answer: Coarse quantization reduces the number of distinguishable intensity/spectral levels, merging different land-cover types that have similar but distinct values, which lowers classification accuracy. Finer quantization preserves these distinctions, improving classification accuracy.

48. A vision system fails in low-light conditions. Explain how image sensing impacts system performance.

Answer: Low light reduces the signal captured by the sensor relative to noise, lowering feature visibility and contrast, which degrades the performance of detection and recognition algorithms. Improving sensor sensitivity, illumination, or applying noise reduction can improve performance.

49. A digital image contains staircase artifacts. Explain how sampling causes this effect.

Answer: Staircase (jaggy) artifacts occur when a low sampling rate is used to represent curved or diagonal edges, causing the continuous boundary to be approximated by a series of discrete pixel steps rather than a smooth line.

50. In an industrial system, explain how acquisition noise affects measurement accuracy.

Answer: Noise introduced during acquisition adds random error to the pixel intensity values used for measurement, causing inaccuracies in dimensions, positions, or other quantities derived from the image. Proper calibration and noise reduction improve measurement accuracy.

51. A camera captures images with motion blur. Explain how image formation and acquisition timing affect clarity.

Answer: During exposure, the image formation process integrates light from a moving object across its changing positions, smearing it across pixels; longer exposure times (acquisition timing) increase this effect. Faster shutter speeds or motion compensation reduce blur and improve clarity.

52. A digital image has poor dynamic range. Explain how quantization influences intensity representation.

Answer: Limited quantization levels compress the range of representable intensities, making it difficult to distinguish between very dark and very bright regions simultaneously, resulting in poor dynamic range. Higher bit-depth quantization allows a wider range of intensities to be accurately represented.

53. In a real-time vision system, explain how acquisition speed impacts processing efficiency.

Answer: Faster acquisition (higher frame rate/quicker sensor readout) reduces latency and enables the system to process and respond to changes in the scene promptly, but excessively short exposure times can reduce image quality/increase noise, so acquisition speed must be balanced against image quality for efficient real-time processing.

54. A system captures images with uneven resolution across regions. Explain how sampling contributes to this issue.

Answer: Non-uniform sampling density, caused by factors such as lens distortion or the geometric arrangement of the sensor array, results in some regions of the scene being sampled more densely than others, producing uneven resolution across the image.

55. In medical imaging, explain how image acquisition affects diagnostic reliability.

Answer: Acquisition parameters such as resolution, contrast settings, sensor calibration, and imaging protocol directly affect the clarity and accuracy with which anatomical structures are captured. Poor acquisition can obscure pathology, reducing the reliability of diagnosis.

56. A digital image appears overly compressed. Explain how sampling and quantization contribute to information loss.

Answer: Compression often involves reducing the sampling rate (downsampling) and/or the number of quantization levels used to represent the image, both of which discard spatial and intensity detail, resulting in a loss of fine information and reduced image quality.

57. A vision system detects objects incorrectly due to poor acquisition. Explain the relationship between sensing and detection accuracy.

Answer: Poor sensing quality, such as noise, low resolution, or poor illumination, degrades the quality of features extracted from the image. Since detection algorithms rely on these features, degraded sensing directly lowers detection accuracy.

58. A high-resolution image still lacks detail. Explain how image formation models influence clarity.

Answer: Even with high resolution, factors modeled by the image formation process, such as defocus, lens distortion, or inadequate optics, can prevent fine detail from being properly captured in the first place. This shows that resolution alone is not sufficient; correct image formation is also essential for clarity.

59. In an automated inspection system, explain how sampling affects defect detection sensitivity.

Answer: A higher sampling rate captures finer spatial variations, enabling the detection of small or subtle defects that would otherwise be missed. A low sampling rate may average out or miss such fine defects, reducing detection sensitivity.

60. A camera captures images with inconsistent intensity levels. Explain how quantization and sensing contribute to this variation.

Answer: Variations in sensor response, illumination, and exposure across different captures affect the raw intensity values sensed, while quantization rounding can add further small inconsistencies. Together, inconsistent sensing conditions and quantization contribute to the variation in recorded intensity levels.

Code of Conduct:

I K.L.V.S Karthik(192424177) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.L.V.S Karthik